

MIT Integration Bee: Problem 20

2026 qualifying round – detailed solution

the problem

evaluate the integral

$$I = \int_0^{\pi/2} \cos^2\left(\frac{\pi}{2} \cos^2\left(\frac{\pi}{2} \cos^2 x\right)\right) dx.$$

the key symmetry

the repeated expression suggests naming its inner transformation. set

$$L = \frac{\pi}{2}, \quad T(t) = L \cos^2 t \quad (0 \leq t \leq L).$$

because $0 \leq \cos^2 t \leq 1$, the transformation maps the interval $[0, L]$ into itself. applying it twice gives

$$T(T(x)) = L \cos^2(L \cos^2 x),$$

so the original integrand is exactly $\cos^2(T(T(x)))$.

reflect a point t across the midpoint of the interval by writing

$$R(t) = L - t.$$

the complementary-angle identity gives the crucial relation

$$\begin{aligned} T(R(t)) &= L \cos^2(L - t) \\ &= L \sin^2 t \\ &= L(1 - \cos^2 t) \\ &= L - T(t) \\ &= R(T(t)). \end{aligned}$$

thus reflecting before applying T has the same effect as reflecting afterward. the relation holds for every $t \in [0, L]$, including $t = T(x)$ because $T(x)$ is also in that interval.

carry the symmetry through both layers

first apply the relation at $t = x$:

$$T(R(x)) = R(T(x)) = L - T(x).$$

now apply T to both sides, and use the same relation again at $t = T(x)$:

$$\begin{aligned} T(T(R(x))) &= T(R(T(x))) \\ &= R(T(T(x))) \\ &= L - T(T(x)). \end{aligned}$$

this is the step that handles both nested cosine transformations; a single complementary-angle identity alone would not yet reach the outermost cosine.

write the full integrand as $F(x) = \cos^2(T(T(x)))$. the reflected value is

$$\begin{aligned} F(R(x)) &= \cos^2(T(T(R(x)))) \\ &= \cos^2(L - T(T(x))) \\ &= \sin^2(T(T(x))) \\ &= 1 - \cos^2(T(T(x))) \\ &= 1 - F(x). \end{aligned}$$

therefore each pair of points x and $L - x$ contributes complementary integrand values:

$$F(x) + F(L - x) = 1.$$

evaluate the integral

use the substitution $u = L - x$. when $x = 0$, $u = L$; when $x = L$, $u = 0$; and $du = -dx$. hence

$$\begin{aligned} I &= \int_0^L F(x) dx \\ &= \int_L^0 F(L - u)(-du) \\ &= \int_0^L F(L - u) du. \end{aligned}$$

rename the dummy variable u as x and add this equality to the original integral:

$$\begin{aligned} 2I &= \int_0^L (F(x) + F(L - x)) dx \\ &= \int_0^L 1 dx \\ &= L = \frac{\pi}{2}. \end{aligned}$$

dividing by two yields

$$\boxed{I = \frac{\pi}{4}}.$$

source: [mit integration bee 2026 qualifying test](#); result checked against the [official answer key](#).